TERTIARY INFOTECH ACADEMY

LAB 01
Build an Observability and Incident Detection Baseline

AWS Certified SysOps Administrator Associate Training
TGS-2024051413 · Version v6.0

Scenario
Northstar Retail has intermittent web latency but no agreed baseline, owner or repeatable incident-evidence process.

Objective
Apply K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.

Deliverable
A CloudWatch dashboard, alarm definition, EventBridge pattern, evidence log and validation screenshots.

Services and tools
CloudWatch · CloudTrail · EventBridge · SNS · AWS CLI · CloudShell

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Open `LAB-PLAN.md`; record the assigned Region, workload tag, notification email and cost boundary before creating resources.
```
open LAB-PLAN.md
```

Evidence note: _____

Step 2 — Confirm the active identity and Region in CloudShell. Save the output in `evidence/caller-identity.txt`.
```
aws sts get-caller-identity | tee evidence/caller-identity.txt
aws configure get region | tee evidence/region.txt
```

Evidence note: _____

Step 3 — Create or select a small tagged EC2 training workload and confirm the `Course=TGS-2024051413` tag is visible.
```
aws ec2 describe-instances --filters 'Name=tag:Course,Values=TGS-2024051413' --query
'Reservations[].Instances[].{Id:InstanceId,State:State.Name,AZ:Placement.AvailabilityZone}'
--output table
```

Evidence note: _____

Step 4 — Replace the placeholder instance ID in `cloudwatch-dashboard.json`, then create the dashboard.
```
aws cloudwatch put-dashboard --dashboard-name SysOps-Lab01 --dashboard-body
file://cloudwatch-dashboard.json
```

Evidence note: _____

Step 5 — Replace the placeholder values in `alarm.json`; create the CPU alarm and verify its current state.

```
aws cloudwatch put-metric-alarm --cli-input-json file://alarm.json
aws cloudwatch describe-alarms --alarm-names SysOps-Lab01-HighCPU
```

Evidence note: _____

Step 6 — Review CloudTrail Event history for your identity and export recent management events.

```
aws cloudtrail lookup-events --max-results 20 --query 'Events[].
{Time:EventTime,Name:EventName,User:Username}' --output table | tee evidence/cloudtrail-
events.txt
```

Evidence note: _____

Step 7 — Create the EventBridge rule from `event-pattern.json`; add the trainer-approved SNS target only after subscription confirmation.

```
aws events put-rule --name SysOps-Lab01-ConsoleChange --event-pattern file://event-
pattern.json --state ENABLED
```

Evidence note: _____

Step 8 — Generate one safe configuration event, locate it in CloudTrail and confirm the EventBridge rule metric changes.

```
aws events describe-rule --name SysOps-Lab01-ConsoleChange | tee evidence/event-rule.txt
```

Evidence note: _____

Step 9 — Complete `INCIDENT-EVIDENCE.md`: separate observation, hypothesis, impact, next test and owner.

```
open INCIDENT-EVIDENCE.md
```

Evidence note: _____

Step 10 — Capture the dashboard, alarm and event rule as JPEG screenshots; compare them with `EXPECTED-ARCHITECTURE.jpg`.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 11 — Run `validate.sh`; resolve missing placeholders or evidence before cleanup.

```
bash validate.sh
```

Evidence note: _____

Step 12 — Delete the lab rule, alarm and dashboard unless the trainer approves reuse; record cleanup evidence.

```
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

Acceptance criteria

- No placeholder values remain
- All resources carry the course tag where supported
- Alarm has an owner and action
- Evidence includes UTC timestamps
- No unapproved paid resource remains

Evidence checklist

□ Dashboard JPEG

□ Alarm JPEG

□ CloudTrail event export

□ EventBridge rule export

□ completed INCIDENT-EVIDENCE.md

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.